

Day 3

Introduction to IP Addressing Theory (1 hour): IPv4 and IPv6: Differences, structure, and importance. **Subnetting basics:** CIDR, subnet masks. **Practical (1 hour):** Assign static IPs to devices and verify connectivity. **Task:** Solve subnetting exercises and calculate valid subnets.

1. List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

ANS :

Feature	IPv4	IPv6
Address Length	32-bit	128-bit
Address Format	Decimal (e.g., 192.168.1.1)	Hexadecimal (e.g., 2001:db8::1)
Number of Addresses	About 4.3 billion	About 340 undecillion
Configuration	Manual or DHCP	Auto-configuration supported
Security	Optional IPsec	Built-in IPsec support
Broadcast Support	Supports Broadcast	No Broadcast, uses Multicast

2. Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

ANS :

Network: 192.168.10.0/24

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

Device	IP Address	Subnet Mask	Default Gateway
Laptop	192.168.10.10	255.255.255.0	192.168.10.1
Phone	192.168.10.20	255.255.255.0	192.168.10.1
Smart TV	192.168.10.30	255.255.255.0	192.168.10.1
Printer	192.168.10.40	255.255.255.0	192.168.10.1

3. Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.

ANS :

CIDR Information

- Network Address: 192.168.1.0
- CIDR: /27
- Subnet Mask: 255.255.255.224

Calculations

- Number of Host Bits = $32 - 27 = 5$
- Hosts per Subnet = $2^5 - 2 = 30$ usable hosts

First Three Valid Subnets

Subnet	Network Address	First Host	Last Host	Broadcast Address
Subnet 1	192.168.1.0	192.168.1.1	192.168.1.30	192.168.1.31
Subnet 2	192.168.1.32	192.168.1.33	192.168.1.62	192.168.1.63
Subnet 3	192.168.1.64	192.168.1.65	192.168.1.94	192.168.1.95

Summary

- Total Subnets = 8
- Usable Hosts per Subnet = 30

4. You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet.

ANS :

Step 1: Convert Subnet Mask to CIDR

- Subnet Mask = 255.255.255.0
- CIDR Notation = /24

Step 2: Determine Network Information

Item	Value
IP Address	10.0.5.17
Subnet Mask	255.255.255.0
CIDR	/24
Network Address	10.0.5.0
Broadcast Address	10.0.5.255
First Valid Host	10.0.5.1
Last Valid Host	10.0.5.254

Result

- Network Address: 10.0.5.0
- Broadcast Address: 10.0.5.255
- Valid Host Range: 10.0.5.1 – 10.0.5.254

5. Use an online subnet calculator tool to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.

ANS :

Subnet Calculator Used: <https://www.subnet-calculator.com/>

Verification Result:

The subnet calculator confirmed that the network address is **10.0.5.0**, the broadcast address is **10.0.5.255**, and the valid host range is **10.0.5.1 to 10.0.5.254**.

IPv4 Subnet Calculator

Result

IP Address:	10.0.5.0
Network Address:	10.0.5.0
Usable Host IP Range:	10.0.5.1 - 10.0.5.254
Broadcast Address:	10.0.5.255
Total Number of Hosts:	256
Number of Usable Hosts:	254
Subnet Mask:	255.255.255.0
Wildcard Mask:	0.0.0.255
Binary Subnet Mask:	11111111.11111111.11111111.00000000
IP Class:	C
CIDR Notation:	/24
IP Type:	Private
Short:	10.0.5.0 /24
Binary ID:	000010100000000000000000101000000000
Integer ID:	167773440
Hex ID:	0xa000500
in-addr.arpa:	0.5.0.10.in-addr.arpa
IPv4 Mapped Address:	::ffff:0a00.0500
6to4 Prefix:	2002:0a00.0500::/48